



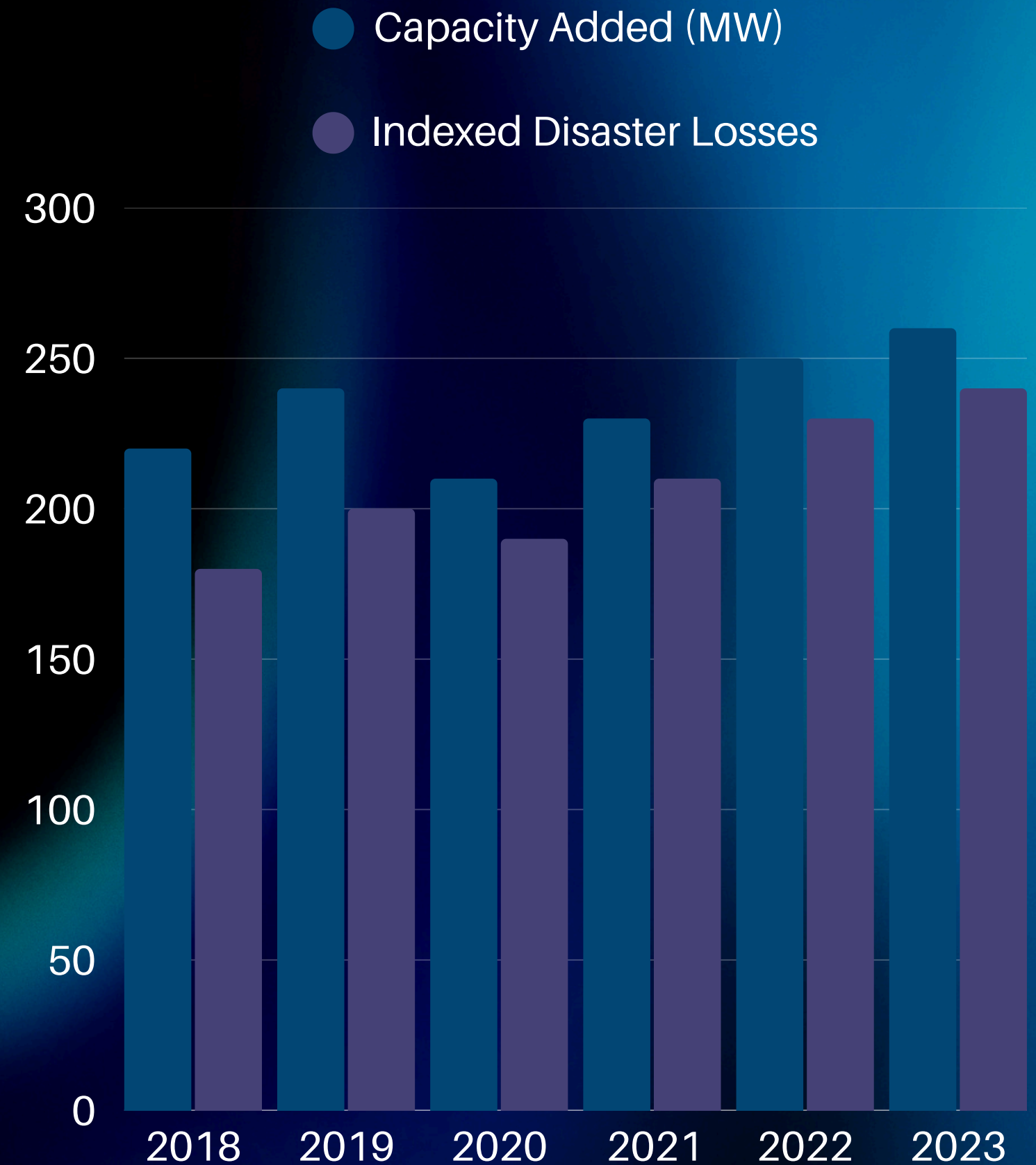
Pricing Risk, Powering Resilience

A 10-Year Energy & Economic Stability
Policy for Uttarakhand (2025-2035)

Uttarakhand's Core Issue

Mispriced Risk, Not Missing Growth

- Uttarakhand does not have a growth problem. It has a risk mispricing problem.
- Every policy failure, cost overruns, landslides, stalled projects, flows from one error: risk is not priced into decisions.
- The state approves projects on capacity added, but pays costs on losses realised.
- Our framework applies Risk-Adjusted Intertemporal State Value (RAISV) to evaluate all major energy investments.





A Resilience-First Growth Model

Uttarakhand's development vision for the next decade is built on a resilience-first model, where growth is pursued not by maximising capacity, but by maximising long-term value after accounting for geological, climatic, and fiscal risk. The state's prosperity depends on building an energy system that is scientifically evaluated, locally resilient, and economically stable.

01

Science-Based Investment

All major energy decisions are evaluated using risk-adjusted metrics such as RAISV, ensuring projects reflect their true long-term cost and vulnerability. Instead of rewarding headline capacity additions, the state shifts to a model where only mathematically viable, risk-priced projects move forward.

02

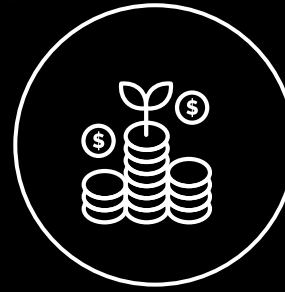
Distributed Resilience Model

The development pathway prioritises decentralised, low-risk, community-owned renewable systems that remain operational even during disasters. By reducing exposure to landslides, floods, and subsidence, the state ensures uninterrupted energy availability while strengthening local livelihoods and climate resilience.

Energy Strategy

Uttarakhand's future energy pathway positions large hydropower not as the primary engine of growth but as a carefully regulated balancing resource whose expansion depends entirely on scientifically priced risk. Instead of approving new dams based on theoretical capacity alone, the state shifts to a model where every project must first demonstrate that its geological, climatic, and financial risks can be fully absorbed by the market—not by taxpayers.

Under this strategy, hydropower is allowed only when risk is transparently priced, externally insured, and economically justified through the RAISV framework, ensuring fiscal discipline and long-term resilience.



Hydropower's Reclassified Role

Hydropower moves from being the default growth driver to a targeted peaking and balancing asset that supports solar and micro-hydro variability. This prevents over-dependence on mega-dams while still leveraging existing hydropower infrastructure efficiently.



"No New Hydro Without Risk Buyers" Rule

Any new large hydropower project must secure 100% disaster-risk insurance through global reinsurance markets, with the full premium paid by the project SPV and embedded into the tariff. This ensures the state undertakes zero fiscal liability, and only economically truthful, risk-adjusted projects survive.



Strengthening Existing Projects

Current hydropower assets continue to operate but under upgraded safety monitoring, real-time geological surveillance, and regular stress audits. This approach protects existing investments while preventing new vulnerabilities from entering the system.

Microgrids > Megaprojects: A Himalayan Energy Architecture

01

Community-Owned Microgrids

Solar + battery microgrids operated by village cooperatives provide reliable, low-cost power that can disconnect from the main grid during emergencies. This ensures uninterrupted electricity even when landslides or floods damage central infrastructure.

02

Micro-Hydro on Minor Streams

Small, low-impact turbines on Himalayan streams generate stable, year-round power without the ecological disruption of large dams. These systems are cheap, fast to deploy, and carry near-zero geological risk.

03

Geothermal Pilot Development

With 40–62 identified hot-spring sites, Uttarakhand begins geothermal exploration to establish a 24/7 baseload resource. This diversifies the energy mix and reduces dependence on river-based systems vulnerable to climate variability.

04

Disaster-Resilient System Design

Distributed assets carry no landslide threat and enable rapid post-disaster recovery. Local ownership, modular infrastructure, and grid-islanding capabilities create a power network that survives and functions even during extreme geological events.



Fiscal Architecture & Risk Pricing

01

Catastrophe-Linked Resilience Bonds

These bonds offer higher yields to investors in normal years, but automatically convert into disaster-relief funding when major events occur. This eliminates sudden fiscal shocks and avoids emergency borrowing after floods, landslides, or infrastructure failures.

02

Geo-Risk Royalty Mechanism

All energy projects pay a variable royalty based on seismic activity, landslide vulnerability, and terrain stability at their site. The collected revenue funds slope stabilization, local disaster preparedness, and livelihood programs in high-risk hill districts, making risky development financially self-correcting.



Uttarakhand adopts a fiscal model where disaster risk is shifted away from taxpayers and towards financial markets and project developers. By using catastrophe-linked instruments and geo-sensitive pricing, the state ensures that only projects able to bear their true geological and climate costs move forward, protecting both the budget and long-term stability.

A Quantitative, Feasible Path to Energy Security

Expected Outcomes:

- Meets projected demand growth without external purchases
- Reduces geological disaster exposure by decentralising assets
- Creates over 50,000 skilled and semi-skilled energy jobs across hill districts

Capacity Targets:

- 5,000 MW new renewable capacity by 2035, composed of:
- 70% micro-hydro (community-owned)
- 30% rooftop and distributed solar
- 10,000 village-level energy cooperatives operational by 2035
- Geothermal exploration and drilling in high-potential belts (Tapovan, Gaurikund, Tapovan–Vishnugad cluster)

Market Strategy



This strategy delivers energy security, fiscal stability, and long-term disaster resilience by grounding every decision in accurate risk pricing. By combining distributed renewables with strictly insured hydropower and innovative financial tools, Uttarakhand achieves sustainable growth without amplifying geological vulnerability.

01

Meeting Energy and Fiscal Constraints

Distributed solar, micro-hydro, and geothermal expansion provide 5,000 MW of new capacity while avoiding costly guarantees or project bailouts. Hydropower is used selectively and only when fully insured, ensuring all expansion fits within the state's limited fiscal space.

02

Ensuring Long-Term Risk Reduction

Microgrids decentralise exposure, geo-risk royalties discourage unsafe locations, and catastrophe-linked bonds shift disaster risk to global investors. Together, these mechanisms protect communities, stabilize budgets, and create a development path that strengthens resilience year after year.



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Our Team

VoltEdge is a multidisciplinary team of engineering students from MNNIT Allahabad, focused on structured problem-solving and clear decision-making. We identify overlooked real-world challenges and translate them into practical, scalable solutions through rigorous analysis and thoughtful design.